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Cost of growth in larval and juvenile African catfish (*Clarias gariepinus*) in relation to growth rate, food intake and oxygen consumption

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Abstract

The effect of fish size on the cost of growth, growth rates, food intake and oxygen consumption was studied in *Clarias gariepinus*. Measurements were performed on larvae reared at 28°C, and results compared to those for juveniles (recalculated from literature). As *C. gariepinus* grows (0.07 mg to 38 g BDW (body dry weight)), the cost of growth increases from 64 to 149 mmol ATP g⁻¹ DW (dry weight) deposited. A higher cost of growth is associated with reduced growth rates (from 141 to 2.4% BDW day⁻¹), and is also reflected in lower gross food conversion efficiencies (93 to 24%). Decreasing growth rates are a result of reductions in relative rates of food intake (152 to 6.2% BDW day⁻¹) and are accompanied by a reduction in oxygen consumption (1.0 to 0.033 μmol O₂ mg⁻¹ DW h⁻¹). Fish size was allometrically related to growth rate, food intake and oxygen consumption. The absence of any distinct breakpoint in these relations suggests a gradual transition from the larval to the juvenile pattern of fish growth and metabolism. Providing that optimal conditions are met (e.g., temperature, oxygen availability and feeding), the pattern of fast growth at low cost might be extended for a longer time span during the transition period from the larval to the juvenile stage, resulting in improved growth and food conversion efficiency. © 1998 Elsevier Science B.V.

Keywords: Feeding and nutrition-fish; Larval rearing; Physiology growth-fish energetics

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1. Introduction

Growth rates (per unit of body mass) decrease as animals increase in size. Increasing body mass gives rise to a diminishing mass-specific oxygen consumption (Oikawa and Itazawa, 1984; Wieser et al., 1988). Growth is an energy demanding process and it accounts for a large portion of the total energy expenditure (Jobling, 1985; Wieser, 1994). Fish larvae grow very rapidly in comparison with older fish. It has been suggested that either the cost of depositing each unit of body mass or the cost involved in maintenance functions are reduced at high growth rates allowing fish larvae to accommodate both the costs of growth and maintenance (Wieser and Medgyesy, 1990; Wieser, 1994). Wieser (1994) suggested that the cost of growth decreases asymptotically as growth rate increases: the cost of growth was estimated to be 66 mmol ATP g⁻¹ dry weight (DW) deposited for animals growing faster than 16.8% body dry weight (BDW) day⁻¹. However, there is no evidence that the cost of growth changes in such a pattern for individuals within a given species growing at different rates.

The cost of growth has been largely attributed to the cost of protein deposition (Jobling, 1985; Brown and Cameron, 1991; Houlihan, 1991). Therefore, the biochemical cost of protein synthesis might be expected to set the lower limit for the cost of growth. When increasing rates of protein synthesis are associated with an increase in protein degradation above its rate at zero growth, the cost of protein deposition (mmol ATP g⁻¹ protein deposited), and thus the cost of growth, will be greater than the cost of protein synthesis (mmol ATP g⁻¹ protein synthesised). However, if the rate of protein degradation remains constant, the cost of protein deposition will be equal to the cost of protein synthesis. Interestingly, the value of 66 mmol ATP g⁻¹ DW deposited estimated by Wieser (1994) is quite close to the 50 mmol ATP g⁻¹ protein synthesised (based on 5 mol ATP needed per peptide bond) estimated to be the minimum biochemical cost of protein synthesis (Reeds et al., 1985).

The primary goal of this study was to investigate the relationship between growth rates and the cost of growth, within a single species. Growth rates, food intake and oxygen consumption were measured in yolk-sac larvae and in fed larvae of the African catfish *Clarias gariepinus* (Burchell), and results compared with values recalculated from published data for juveniles of the same species (Hogendoorn, 1983). The relationships between growth rates, estimated costs of growth and food conversion efficiencies at different fish sizes were analysed, and the findings are discussed in the perspective of growth optimisation in fish culture operations.

2. Materials and methods

2.1. Fish

Eggs of the African catfish, *C. gariepinus*, were obtained by artificial fertilisation (Hogendoorn and Vismans, 1980) of eggs obtained from broodstock held under conditions described by Richter et al. (1995). Eggs produced by two females (individual wet weight ranging from 0.70 to 1.58 kg) were mixed with the sperm from two males (0.72 to 1.96 kg) and were incubated in a recirculation system at 28.0 ± 0.2°C (daily

fluctuation). Offspring originating from one batch of eggs were used for measurements during the yolk-sac stage. Growth and oxygen consumption of exogenously fed larvae were measured on offspring from a second egg batch. Offspring from three additional batches of eggs were used for estimation of consumption of *Artemia* during larval development.

2.2. Rearing, sampling and body weight

Yolk-sac larvae hatched about 22 h after fertilisation, and were reared at $28.0 \pm 0.2^\circ\text{C}$ in 16 l aquaria until complete yolk absorption. The aquaria had a continuous water flow (0.2 to 0.5 l min^{-1}) from a recirculation system, and gentle aeration. The larvae were not fed. Larvae were sampled at 26, 45, 59, 72 and 84 h post-fertilisation for determination of dry matter content. At each sampling point, three samples of 50–150 individuals were taken both for whole larvae and for larval bodies (whole larvae minus yolk-sac). Samples were stored at -20°C until dry matter determination. Samples were freeze dried, and dry weights determined ($\pm 10 \mu\text{g}$) using a Mettler balance (AE160).

For the experiments conducted on fed larvae, groups of 1200 larvae were transferred to each of three 16-l aquaria when yolk absorption was almost complete (48 h after hatching). Larvae were reared in a recirculation system, with 5 to 10% water renewal per day and UV filtration. The aquaria had a flow rate of 1 to 1.5 l min^{-1} , and were kept at $28.0 \pm 0.3^\circ\text{C}$ with the water continuously aerated. During 10 days, the larvae were fed *Artemia* nauplii five times a day (equal portions) at intervals of 3.5 h between 08.30 and 22.30 h. Daily ration was calculated to be near satiation according to Verreth and den Bieman (1987), allowing for maximum growth. *Artemia* cysts (strain Great Salt Lake, Sanders) were incubated for 24 h in 35 ‰ salt water at 28°C . Newly hatched nauplii were stored in salt water at 4°C . A sample of 10 to 50 larvae was taken from each of the three aquaria for dry matter determination at 0, 2, 3, 4, 5, 6, 8, and 9 days of exogenous feeding, before the first daily meal. Samples were stored at -20°C until analysed. Larvae were dried overnight at 70°C , then kept at 103°C for 4 h, and DW measured ($\pm 10 \mu\text{g}$) with a Mettler balance (AE160).

We used BDW to estimate growth rates and mass-specific rates of oxygen consumption. Relative growth rates (RGR, %BDW day^{-1}) were calculated as (Ricker, 1958):

$$\text{RGR} = (e^g - 1) \cdot 100 \text{ with } g = (\ln \text{BDW}_2 - \ln \text{BDW}_1) / (t_2 - t_1)$$

where BDW_2 and BDW_1 are the body DW in samples taken at t_2 and t_1 .

2.3. Food intake measurements

Food intake (FI, %BDW day^{-1}) for yolk-sac larvae was estimated based on the absorption of yolk material between two sampling points:

$$\text{FI} = (\text{YDW}_1 - \text{YDW}_2) / [(\text{BDW}_1 + \text{BDW}_2) / 2] \cdot (t_2 - t_1) \cdot 100$$

where BDW_2 , BDW_1 , YDW_1 and YDW_2 are the body DW and yolk DW (mg/individual) at two samples times (t_2 and t_1). Yolk DW was calculated as the difference between the whole larvae DW and the body DW at each sampling point.

Food intake of fed larvae was estimated on days 2 and 8. Decapsulated *Artemia* cysts were hatched as previously mentioned, but they were held in salt water containing L-[2,6-³H]phenylalanine at a specific activity of 2.04 Tbq mmol⁻¹ phenylalanine, to give a final concentration of 12.6 kbq ml⁻¹. During hydration, *Artemia* cysts take up considerable amounts of dissolved amino acids (D'Agostino, 1980). Newly hatched nauplii were fed to groups of approximately 400 (day 2) or 180 (day 8) larvae held in bottles containing 200 ml of continuously aerated tap water. The bottles were kept in a water bath at 28.0 ± 0.1°C. After 30 min, six samples of five larvae (day 2) or two larvae (day 8) were taken by siphoning and were rinsed well with tap water. Satiation time for *C. gariepinus* fed *Artemia* is about 30 min, and within that period, no material is defecated (Haylor, 1993). The experiment was repeated three times, always at the first meal of the day. Samples were placed into scintillation vials to which 1 ml tissue solubiliser was added (Soluene 100, Packard). Vials were incubated for 2 h at 50°C to aid solubilisation. After cooling, 4 ml scintillation liquid (Hionic Fluor, Packard) was added and samples were counted for radioactivity (dpm, disintegrations per minute). Samples of newly hatched *Artemia* were also prepared in the same manner and counted. Food intake (FI, %BDW day⁻¹) was estimated as:

$$FI = [(R_{\text{larva}}/SR_{\text{Artemia}})/BDW_{\text{larva}}] \cdot 5 \cdot 100.$$

Where R_{larva} is the total radioactivity per larva (dpm larva⁻¹), SR_{Artemia} is the specific radioactivity in *Artemia* samples (dpm mg⁻¹ DW), BDW_{larva} the body DW (mg DW larva⁻¹) and 5 the number of meals in one day. Although we only measured the first meal of the day, we assumed that the food intake in the 5 meals of the day was identical.

2.4. Oxygen consumption measurements

Oxygen consumption was measured using a flow-through respirometer. A constant flow (12 ml min⁻¹) of 28.0 ± 0.1°C tap water was pumped through 150 ml chambers containing the larvae. The flow was maintained by a peristaltic pump, and measured gravimetrically every hour after collection of outlet water for 1 min. One chamber without fish was always used as a blank and as a reference for the inlet water in the other chambers. The oxygen electrode (WTW TriOxmatic 300 sensor) received a continuous water flow alternating from chamber to chamber at intervals of 6 min. This alternating water flow was automatically regulated by magnetic valves connected to a timer. Oxygen concentrations were measured by a precision oxygen meter (WTW Microprocessor-Precision-Oxygen Meter Oxi 3000, Weilheim, Germany). Oxygen concentration in outlet water was never less than 70% of air saturation.

Oxygen consumption of yolk-sac larvae, was monitored in three chambers containing 250 to 300 larvae each. Measurements were taken from hatching until complete yolk absorption. For the fed larvae, oxygen consumption was measured on days 2, 5, and 8 of exogenous feeding. Measurements were performed at the time of the first meal of the day (about 10 h after the preceding meal), both in larvae fed a single meal (fed group) and in unfed larvae (fasted group). The fed group was allowed to feed for 30 min in the growing tanks, and rapidly transferred to the measurement chambers. Two chambers containing 41 to 213 larvae were used for each group. Measurements of oxygen consumption were taken for 8 h, and no food was given during the measurement period.

To estimate the daily oxygen consumption of fed larvae, only the first 3.5 h after the meal were considered, and extrapolated to the following four meals.

Oxygen consumption (MO_2 , $\text{nmol O}_2 \text{ mg}^{-1} \text{ DW h}^{-1}$) was calculated as:

$$\text{MO}_2 = ({}_b\text{O}_2 - {}_c\text{O}_2) \cdot \text{Flow} / (n \cdot \text{BDW} \cdot 32) \cdot 10^6$$

where ${}_b\text{O}_2$ and ${}_c\text{O}_2$ are the oxygen concentrations (mg l^{-1}) in the outlet of the blank chamber and of a chamber with larvae, respectively, Flow is the flow through the chambers (l h^{-1}), n is the number of larvae in the chamber, BDW the mean body DW (mg), 32 the molecular weight of oxygen, and 10^6 is a factor to correct for units.

2.5. Estimation of the cost of growth

The cost of growth was estimated assuming that total oxygen consumption (MO_2 , $\text{nmol O}_2 \text{ mg}^{-1} \text{ DW day}^{-1}$) is composed of two parts: the energy needed for maintenance and the energy required for growth and associated processes (Smith, 1957; Rombough, 1994; Wieser, 1994). This can be formalised as:

$$\text{MO}_2 = \text{COM} + \text{COG} \cdot \text{RGR}$$

where COM ($\text{nmol O}_2 \text{ mg}^{-1} \text{ DW day}^{-1}$) represents the cost of maintenance, COG ($\text{nmol O}_2 \text{ mg}^{-1} \text{ DW deposited}$) the cost of growth, and RGR ($\% \text{BDW day}^{-1}$) the relative growth rate.

Costs of growth (COG, $\text{mmol ATP g}^{-1} \text{ DW deposited}$) for yolk-sac larvae were calculated by regression of mass-specific oxygen consumption (MO_2 , $\text{nmol O}_2 \text{ mg}^{-1} \text{ DW day}^{-1}$) on relative growth rate (RGR, $\% \text{BDW day}^{-1}$). The slope of this equation estimates the cost of growth (Rombough, 1994; Wieser, 1994) provided that differences in units are taken into account. The unit ' $(\text{nmol O}_2 \text{ mg}^{-1} \text{ DW day}^{-1}) / (\% \text{BDW day}^{-1})$ ' can be transformed to ' $\text{mmol ATP g}^{-1} \text{ DW deposited}$ ' if 1 mol of oxygen consumed is taken to be equivalent to the synthesis of 6 mol ATP (Reeds et al., 1985). The intercept of the regression provides an estimate for the metabolic rate at zero growth (cost of maintenance). Also, units here can be converted to ' $\text{mmol ATP g}^{-1} \text{ DW deposited day}^{-1}$ '.

For fed larvae, the cost of growth was estimated by subtracting the oxygen consumption of fed larvae from the oxygen consumption of fasted larvae, and divide it by RGR. Metabolism of the fasted larvae was assumed to be indicative of maintenance metabolism. Oxygen consumption of fed larvae (${}_{\text{fed}}\text{MO}_2$, $\text{nmol O}_2 \text{ mg}^{-1} \text{ DW day}^{-1}$) was calculated as:

$${}_{\text{fed}}\text{MO}_2 = 5 \cdot \int_{\text{fed}}^{3.5} \text{MO}_2 + \int_{\text{fast}}^{6.5} \text{MO}_2$$

where 5 is the number of meals in a day, $\int_{\text{fed}}^{3.5} \text{MO}_2$, the cumulative oxygen consumption in the first 3.5 h after a meal in fed larvae, and $\int_{\text{fast}}^{6.5} \text{MO}_2$ is the cumulative oxygen consumption in the 6.5 h after start of the measurements in the fasted group. The interval between meals was 3.5 h, and it was assumed that oxygen consumption in the five meals of the day is identical. Similarly, oxygen consumption for unfed larvae (${}_{\text{fast}}\text{MO}_2$, $\text{nmol O}_2 \text{ mg}^{-1} \text{ DW day}^{-1}$) was calculated as:

$${}_{\text{fast}}\text{MO}_2 = 4 \cdot \int_{\text{fast}}^6 \text{MO}_2.$$

Costs of growth (COG, mmol ATP g⁻¹ DW deposited) were then estimated as:

$$\text{COG} = (\text{fed} \cdot \text{MO}_2 - \text{fast} \cdot \text{MO}_2) / \text{RGR} \cdot 6$$

where 6 is the factor to convert moles of oxygen consumed to moles of ATP synthesised (Reeds et al., 1985).

Costs of growth for juveniles of *C. gariepinus* (data from Hogendoorn, 1983) were calculated as for fed larvae, using oxygen consumption measurements of growing fish and fish at maintenance.

2.6. Literature experiments

The data presented for *C. gariepinus* juveniles by Hogendoorn (1983) were analysed. In that study, growth, food intake, and oxygen consumption were measured for four size groups of *C. gariepinus*, of initial sizes of approximately 1.5, 7.5, 45 and 95 g fresh weight. Fish were reared for 21 to 42 days at 25°C. For each size group, fish were reared at five feeding levels, i.e., deprivation, maintenance (zero growth), optimal (maximum conversion efficiency), and two and three times optimal, as estimated by preliminary results. In the present study, only the data referring to fish growing at maximum rates in each of the size groups are presented. Oxygen consumption data of fish at maintenance were also used for estimation of the cost of growth.

2.7. Statistics

Student's *t*-tests (Proc TTEST), and linear regressions (Proc REG) were performed with the SAS Statistical software package. To calculate allometric relations, a logarithmic transformation was applied to the data.

3. Results

Weight gain, food intake, oxygen consumption, and gross food conversion efficiencies (K_1 , mg growth/mg food intake $\times$ 100) of the fish of different sizes are shown in Table 1. Yolk-sac larvae had higher food intakes, growth rates and (mass-specific) rates of oxygen consumption than both fed larvae and juvenile *C. gariepinus*. Highly significant correlations were found between BDW (mg) and relative growth rate (RGR, %BDW day⁻¹), food intake (FI, %BDW day⁻¹) and mass-specific oxygen consumption (MO₂, nmol O₂ mg⁻¹ DW h⁻¹):

$$\begin{aligned} \text{RGR} &= 57.62 \text{ BDW}^{-0.276} & (r^2 = 0.95, n = 10, P < 0.001) \\ \text{FI} &= 70.76 \text{ BDW}^{-0.185} & (r^2 = 0.91, n = 8, P < 0.001) \\ \text{MO}_2 &= 469.8 \text{ BDW}^{-0.236} & (r^2 = 0.99, n = 10, P < 0.001) \end{aligned}$$

These regressions refer to a range of BDW from 70 μ g to 37.7 g. Only the first three sampling points in the yolk-sac stage were considered. The latter two were not included since yolk supply was limiting growth (Conceição et al., 1993). From the data on juveniles (Hogendoorn, 1983), only the groups growing at maximum rates were used in the allometric relations.

Table 1

The mass-specific oxygen consumption (MO_2), food intake (FI), relative growth rate (RGR) and gross conversion efficiency (K_1) at different body DW (BDW) of *C. gariepinus*

Stage	Age ^a	BDW (mg)	MO_2 nmol O_2 mg^{-1} DW (h^{-1})	FI (%BDW day^{-1})	RGR (%BDW day^{-1})	K_1 (%)
Yolk-sac larvae	35 hpf	0.07 (0.003)	990.6 (37.8)	152.1	141.0	92.7
	45 hpf	0.12 (0.009)	823.9 (54.9)	—	124.7	—
	59 hpf	0.18 (0.005)	722.1 (38.2)	95.0	78.1	82.2
	72 hpf	0.26 (0.012)	581.2 (35.2)	73.0	69.4	95.2
	84 hpf	0.31 (0.002)	464.6 (29.9)	38.4	28.2	73.9
Fed larvae	2 def	0.56 (0.005)	438.6 (46.7)	51.5 (2.8)	36.4	70.6
	5 def	1.77 (0.097)	341.8 (16.1)	—	67.2	—
	8 def	5.02 (0.030)	317.4 (3.55)	51.5 (1.0)	40.8	79.3
Juveniles ^b		851	106.9	27.9	8.6	30.8
		6449	57.7	14.8	7.6	51.1
		17833	54.1	15.7	3.7	23.8
		37685	32.8	6.2	2.4	37.9

Standard deviations (pooled samples) are given in parentheses.

$n = 3$ For BDW and FI; $n = 2$ for MO_2 .

^aAge is given in hours post-fertilisation (hpf) or days of exogenous feeding (def).

^bRecalculated from Hogendoorn (1983). MO_2 values were adjusted from 25 to 28°C using a Q_{10} of 2.0 (Machiels and Henken, 1986).

As *C. gariepinus* got bigger, the relative growth rate decreased faster than food intake and this was reflected in a decrease in K_1 (Table 1). Food conversion efficiencies were approximately 90, 65, and 30% for yolk-sac larvae, fed larvae and juvenile fish, respectively.

Estimates for the cost of growth and the cost of maintenance at different sizes for the

Table 2

The costs of growth (COG) and cost of maintenance (COM) at different body DW (BDW) of *C. gariepinus*

Stage	Age ^a	BDW (mg)	COM ($\mu\text{mol ATP mg}^{-1}$ DW h^{-1})	COG (mmol ATP g^{-1} DW)
Yolk-sac larvae		0.07 to 0.31	1.964 (0.346)	63.5 (9.2)
Fed larvae	2 def	0.56	1.744 (0.099)	56.1
	5 def	1.77	1.264 (0.058)	51.3
	8 def	5.02	1.202 (0.088)	49.4
Juveniles ^b		851	0.364	148.9
		6449	0.238	95.8
		17833	0.148	121.0
		37685	0.121	94.4

Standard deviations (pooled samples) are given in parentheses.

$n = 2$ For COM of fed larvae.

^aAge is given in days of exogenous feeding (def).

^bRecalculated from Hogendoorn (1983). COM values were adjusted from 25 to 28°C using a Q_{10} of 2.0 (Machiels and Henken, 1986).

three stages are given in Table 2. The cost of growth of yolk-sac larvae was calculated as 63.5 mmol ATP g⁻¹ DW deposited from the relation:

$$\text{MO}_2 = 327.3 + 4.407 \cdot \text{RGR} \quad (r^2 = 0.94, n = 5, P < 0.001).$$

For larvae fed on *Artemia*, the cost of growth was estimated as 56.1 (day 2), 51.5 (day 5) and 49.4 (day 8) mmol ATP g⁻¹ DW deposited, based on the rates of oxygen consumption of Fig. 1. Costs of growth estimated for juveniles growing at maximum rates ranged from 94.4 to 148.9 mmol ATP g⁻¹ DW deposited.

Concerning the cost of maintenance, two distinct phases during ontogeny can be recognised. The difference between the cost of maintenance in yolk-sac larvae and in larvae on day 2 of exogenous feeding was not significant (*t*-test, *P* > 0.05). After day 2, the cost of maintenance (COM, μmol ATP mg⁻¹ DW h⁻¹) decreased. COM of juvenile fish were calculated by adjusting measured values (Hogendoorn, 1983) to 28°C using a

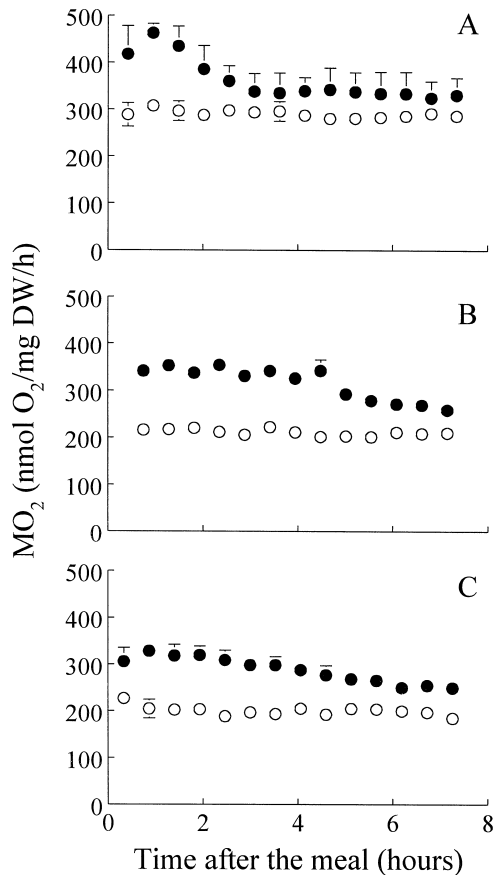


Fig. 1. The mass-specific oxygen consumption (MO₂) in larvae of *C. gariepinus* after being fed a single meal of *Artemia* nauplii at (A) day 2, (B) day 5 and (C) day 8 of exogenous feeding. Values are given for fed (●) and fasted larvae (○). Error bars are standard deviations of two measurements in groups of larvae.

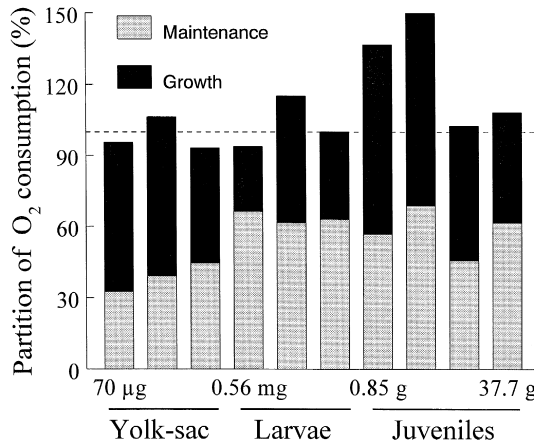


Fig. 2. The partition of oxygen consumption between maintenance (grey bars) and growth (black bars) at the different larval sizes of *C. gariepinus*. Based on the COM and COG estimates from Table 2.

Q_{10} of 2.0 (Machiels and Henken, 1986). From day 2 of exogenous feeding onwards, the following allometric relation was obtained:

$$\text{COM} = 1.586 \cdot \text{BDW}^{-0.211} \quad (r^2 = 0.98, n = 7, P < 0.001).$$

Based on the cost estimates given in Table 2, the partitioning of the total oxygen consumption between growth and maintenance was predicted (Fig. 2). The two smaller juvenile size groups suggest that the fish would need considerably more oxygen than that measured to satisfy maintenance and growth. Yolk-sac larvae seem to allocate more energy to growth than older fish.

4. Discussion

Larvae of *C. gariepinus* have high growth rates (36.4 to 141% BDW day^{-1}) and high food conversion efficiencies (70.6 to 95.2%, dry matter basis) when compared with the larvae of other fish species (Houde, 1989; Wieser and Medgyesy, 1990; Keckeis and Schiemer, 1992; Rombough, 1994), in which growth rates range from 1 to 35% BDW day^{-1} , and conversion efficiencies from 10 to 66%. Growth rates (2.4 to 8.6% BDW day^{-1}) and food conversion efficiencies (23.8 to 51.1%, dry matter basis) of juvenile *C. gariepinus* are comparable to those of juveniles of other fish species. Bowen (1987) listed a series of species growing close to maximum rates, where growth rates ranged from 1.2 to 6.6% BDW day^{-1} , and food conversion efficiencies (protein basis) ranged from 21 to 48%.

Accurate estimation of the cost of growth relies on the assumption that the costs of growth and maintenance are additive, i.e., that both the cost of maintenance will not change with different sizes and/or feeding levels, and the cost of growth is constant (Rombough, 1994). In at least some of the groups of juvenile fish, the cost of maintenance may have been reduced with increasing feeding level to accommodate the

cost of growth. This is suggested by the sums of the costs of maintenance and growth exceeding 100% in some groups (Fig. 2), and also by the considerable variation in the estimates of the cost of growth (Table 2). Such a reduction in the cost of maintenance to accommodate the cost of growth has been suggested before (Kiørboe, 1989; Wieser and Medgyesy, 1990; Rombough, 1994; Wieser, 1994). If this happens in juvenile *C. gariepinus*, then at least some of the estimates for the cost of growth in the present study are underestimated. The estimates for the cost of growth in fed larvae were obtained by extrapolating to five meals per day based on the postprandial oxygen consumption of one meal. This could bring some bias to the estimates if the different meals were of different sizes. However, the measurements were done after the first meal of the day which would tend to be the larger (so would oxygen consumption), as the larvae were unfed for 10 h. Thus, if an error exists, it should be an overestimation of the cost of growth, which seems unlikely as these values are already close to the minimum theoretical value (see below).

In larval *C. gariepinus*, the estimates for the cost of growth (49.4 to 63.5 mmol ATP g⁻¹ DW deposited) were close to the minimum biochemical cost of protein synthesis (50 mmol ATP g⁻¹ protein synthesised). A comparable low cost of growth has been found for herring embryos and larvae (Kiørboe and Møhlenberg, 1987; Kiørboe et al., 1987) and turbot larvae growing between 15.4 and 22.8% BDW day⁻¹ (Conceição et al., unpublished data). The cost of protein synthesis was also close to the biochemical minimum in yolk-sac larvae of *C. gariepinus* (Conceição et al., 1997) and in fast growing juvenile tilapia (Houlihan et al., 1993). However, in juvenile *C. gariepinus* growing at maximum rates, the cost of growth seems to be at least two to three times the minimum biochemical cost of protein synthesis. Fish larvae growing at slower rates (Wieser and Medgyesy, 1990; Houlihan et al., 1992, 1995b) have a cost of growth several fold higher than this minimum theoretical value. Wieser (1994) based on literature data for a wide range of species proposed that in general the cost of growth in aquatic ectotherms would be about three times the biochemical cost of protein synthesis. The present study shows that *C. gariepinus* has a variable cost of growth, the cost of growth being reduced at high growth rates. We recalculated data for Japanese whiting, *Sillago japonica* (Oozeki and Hirano, 1994) and estimated that the cost of growth was 60 mmol ATP g⁻¹ DW deposited for larvae growing at 34% BDW day⁻¹, while in larvae growing at 18% BDW day⁻¹ the cost of growth was 134 mmol ATP g⁻¹ DW deposited. Thus, it seems that individual species follow the pattern that was previously suggested based on an inter-specific comparison (Wieser, 1994). The cost of growth is variable, decreasing towards the minimum biochemical cost of protein synthesis at high growth rates.

5. Conclusion

It is hypothesised that the concurrent high rates of protein synthesis and not the high growth rates are the cause of the reduced cost of growth. The cost of protein synthesis was shown to decrease with augmenting rates of protein synthesis in larval nase (*Chondrostoma nasus*) (Houlihan et al., 1992), in trout hepatocytes (Pannevis and Houlihan, 1992) and in different cell lines (Smith and Houlihan, 1995). It has been

proposed that this could be due to the existence of a fixed component of this cost, independent of the rate of protein synthesis (Pannevis and Houlihan, 1992). Houlihan et al. (1995a) suggested that a constant activation of tRNA and a constant production of rRNA would contribute to this fixed energy expenditure. At low growth rates this fixed component of the cost of protein synthesis would have a large contribution to the cost of growth when compared with the cost of peptide-bond formation.

High growth rates in fish larvae seem to be associated with a low cost of growth and high food conversion efficiencies. The present paper suggests that the transition from larval to juvenile pattern of fish growth and metabolism is a gradual process. If growth rates reach levels that allow for a reduced cost of growth, food conversion efficiencies will increase, allowing for higher growth. Thus, higher growth may bring an amplification effect on growth itself. Once fish is reared under optimal conditions, e.g., oxygen levels, feeding regime, diet quality and water temperature, the larval pattern of high growth rates/low cost of growth/high food conversion efficiencies might be extended for a longer time span in the larval/juvenile transition period. Conversely, sub-optimal conditions will probably lead to increasingly restrained growth rates and food conversion efficiencies.

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